

Biology Quick Facts & Formula Book

Important ratios, facts and rapid revision

Cell Biology Quick Facts

- Cell theory: cells are the basic structural and functional units of living organisms.
- Prokaryotic cells lack a membrane-bound nucleus; eukaryotic cells possess a membrane-bound nucleus.
- DNA stores hereditary information; RNA participates in information transfer and protein synthesis.

Genetics

- Mendelian monohybrid phenotypic ratio in complete dominance: 3:1.
- Mendelian monohybrid genotypic ratio: 1:2:1.
- Dihybrid phenotypic ratio under independent assortment and complete dominance: 9:3:3:1.
- Probability of independent events occurring together is obtained by multiplying their probabilities.

Molecular Biology

- DNA base pairing: A pairs with T; G pairs with C.
- During transcription, RNA is synthesized using a DNA template.
- During translation, codons on mRNA specify amino acids through the genetic code.

Ecology

- Energy transfer between trophic levels is inefficient; only a fraction of energy passes to the next level.
- Population density can be expressed as the number of individuals per unit area or volume.
- Species diversity considers both richness and relative abundance.

Human Physiology

- Heart circulation involves pulmonary and systemic circuits.
- Gas exchange occurs across thin respiratory surfaces by diffusion.
- Nephron is the structural and functional unit of the kidney.

Photosynthesis & Respiration

- Photosynthesis converts light energy into chemical energy and ultimately forms carbohydrates from carbon dioxide and water.
- Cellular respiration releases usable energy from organic molecules through a sequence of reactions.
- ATP acts as a major energy-transfer molecule in cells.

Exam Reminder

- Biology often tests exact terminology, examples, sequences and diagram labels.

- Do not confuse similarly named structures or hormones; revise comparison tables.
- Use NCERT-style wording as the primary revision language for NEET-oriented preparation.